

KISSsoft - Release 03/2017

Team-SolidSQUAD

Файл

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АНАЛИЗ РЕЗУЛЬТАТОВ(Оценка важных свойств)

Комментарий:

№ = номер варианта
diff_i = отклонение от заданного передаточного отношения в %
кг = вес в кг
слайд = удельное скольжение (максимальное значение)
v.Slide = Скорость скольжения (м/с, максимальное значение)
AC/AE = Рабочий участок линии зацепления на входе AC / рабочий участок линии зацепления AE (характеристики при трении)
del_cg = Стандартное отклонение жесткости при обкатке (Н/мм/мум)
1-eta = Потери в % (1.0-общий КПД)
Safety = запас прочности (ножка зуба и боковая поверхность, 0 = высоко, 1 = достаточно, 2 = низко)
(SF-min: 0.60/ 1.20/ 1.40 SH-min: 0.60/ 0.90/ 1.00)
Summary = Общая оценка (оценочно)
(50.0%:del_cg 20.0%:diff_i 100.0%:kg 35.0%:Slide 0.0%:v.Slide
0.0%:AC/AE 10.0%:1-eta 100.0%:Safety)

(Для таблицы справедливо: Чем меньше число, тем лучше!)

No.	diff_i	kg	Slide	v.Slide	AC/AE	del_cg	1-eta	Safety	Summary
1	-1.389	10.008	1.399	0.757	0.506	0.000	0.982	0.648	0.325
2	-1.389	9.974	1.133	0.800	0.460	0.000	0.982	0.655	0.327
3	-1.389	9.940	0.912	0.860	0.412	0.000	1.010	0.661	0.327
4	0.000	10.037	1.757	0.796	0.494	0.000	1.065	0.660	0.328
5	0.000	10.002	1.379	0.858	0.448	0.000	1.071	0.667	0.328
6	0.000	9.968	1.081	0.919	0.401	0.000	1.106	0.673	0.329

7	-4.762	9.888	2.132	0.853	0.488	0.000	1.080	0.638	0.336
8	0.000	10.208	1.480	0.761	0.502	0.000	1.001	0.649	0.323
9	0.000	10.173	1.190	0.815	0.456	0.000	1.003	0.656	0.324
10	0.000	10.139	0.952	0.875	0.409	0.000	1.033	0.662	0.324
11	1.389	10.236	1.868	0.809	0.490	0.000	1.086	0.664	0.335
12	1.389	10.201	1.450	0.871	0.445	0.000	1.093	0.670	0.336
13	1.389	10.167	1.127	0.932	0.397	0.000	1.130	0.677	0.336
14	-4.762	10.056	1.753	0.815	0.496	0.000	1.014	0.602	0.320
15	-4.762	10.022	1.381	0.879	0.450	0.000	1.018	0.627	0.328
16	-4.762	9.988	1.087	0.942	0.403	0.000	1.050	0.633	0.328
17	0.000	10.311	0.839	0.829	0.417	0.000	0.967	0.656	0.322
18	1.389	10.409	1.565	0.768	0.499	0.000	1.020	0.652	0.330
19	1.389	10.374	1.248	0.829	0.453	0.000	1.023	0.658	0.330
20	1.389	10.339	0.992	0.890	0.406	0.000	1.056	0.664	0.331
21	2.778	10.438	1.989	0.821	0.487	0.000	1.106	0.668	0.343
22	2.778	10.403	1.525	0.883	0.441	0.000	1.114	0.675	0.343
23	-4.762	10.225	1.475	0.787	0.504	0.000	0.952	0.583	0.311
24	-4.762	10.191	1.190	0.837	0.458	0.000	0.953	0.592	0.313
25	-4.762	10.157	0.956	0.899	0.411	0.000	0.980	0.605	0.317
26	-3.439	10.255	1.863	0.827	0.493	0.000	1.033	0.619	0.324
27	-3.439	10.221	1.452	0.891	0.447	0.000	1.038	0.630	0.326
28	-3.439	10.187	1.132	0.954	0.400	0.000	1.073	0.637	0.327
29	1.389	10.549	1.084	0.785	0.461	0.000	0.961	0.650	0.327
30	1.389	10.513	0.876	0.845	0.413	0.000	0.988	0.657	0.328
31	2.778	10.613	1.658	0.782	0.495	0.000	1.040	0.654	0.337
32	2.778	10.577	1.310	0.843	0.449	0.000	1.045	0.660	0.337
33	2.778	10.542	1.033	0.903	0.402	0.000	1.079	0.666	0.338
34	4.167	10.642	2.122	0.832	0.484	0.000	1.126	0.673	0.352
35	-4.762	10.362	1.035	0.792	0.466	0.000	0.894	0.584	0.309
36	-4.762	10.328	0.844	0.853	0.419	0.000	0.917	0.592	0.312
37	-3.439	10.426	1.561	0.790	0.501	0.000	0.971	0.584	0.309
38	-3.439	10.392	1.249	0.851	0.455	0.000	0.973	0.595	0.311
39	-3.439	10.358	0.996	0.913	0.408	0.000	1.002	0.612	0.316
40	-2.116	10.457	1.983	0.839	0.490	0.000	1.052	0.628	0.325
41	2.778	10.790	1.408	0.751	0.504	0.000	0.979	0.643	0.331
42	2.778	10.754	1.137	0.800	0.458	0.000	0.980	0.651	0.333
43	2.778	10.718	0.913	0.860	0.410	0.000	1.009	0.657	0.334

44	4.167	10.819	1.758	0.795	0.492	0.000	1.060	0.656	0.344
45	4.167	10.782	1.375	0.856	0.446	0.000	1.066	0.663	0.344
46	4.167	10.746	1.075	0.917	0.399	0.000	1.102	0.669	0.345
47	-3.439	10.599	1.330	0.773	0.509	0.000	0.913	0.576	0.305
48	-3.439	10.565	1.085	0.807	0.463	0.000	0.912	0.584	0.306
49	-3.439	10.530	0.880	0.868	0.416	0.000	0.937	0.593	0.309
50	-2.116	10.630	1.652	0.801	0.498	0.000	0.989	0.588	0.307
51	-2.116	10.595	1.310	0.864	0.452	0.000	0.993	0.600	0.310
52	-2.116	10.560	1.037	0.926	0.405	0.000	1.024	0.625	0.319
53	4.167	10.997	1.487	0.755	0.500	0.000	0.997	0.644	0.338
54	4.167	10.961	1.191	0.815	0.454	0.000	1.000	0.652	0.339
55	4.167	10.924	0.951	0.874	0.407	0.000	1.031	0.658	0.340
56	-2.116	10.805	1.403	0.777	0.506	0.000	0.931	0.576	0.302
57	-2.116	10.769	1.136	0.822	0.460	0.000	0.931	0.584	0.303
58	-2.116	10.734	0.916	0.883	0.412	0.000	0.957	0.594	0.306
59	-0.794	10.835	1.750	0.813	0.494	0.000	1.008	0.592	0.306
60	-0.794	10.800	1.375	0.876	0.448	0.000	1.013	0.609	0.311
61	-0.794	10.764	1.079	0.938	0.401	0.000	1.046	0.628	0.317
62	4.167	11.104	0.842	0.829	0.414	0.000	0.966	0.653	0.338
63	-2.116	10.910	0.811	0.838	0.420	0.000	0.897	0.587	0.303
64	-0.794	11.012	1.480	0.780	0.502	0.000	0.948	0.577	0.299
65	-0.794	10.976	1.190	0.836	0.456	0.000	0.950	0.586	0.301
66	-0.794	10.940	0.953	0.897	0.409	0.000	0.978	0.596	0.304
67	0.529	11.043	1.857	0.825	0.491	0.000	1.027	0.600	0.310
68	0.529	11.007	1.443	0.888	0.445	0.000	1.032	0.626	0.318
69	-0.794	11.191	1.270	0.763	0.511	0.000	0.893	0.572	0.297
70	-0.794	11.154	1.039	0.792	0.464	0.000	0.892	0.579	0.298
71	-0.794	11.118	0.845	0.853	0.417	0.000	0.916	0.587	0.300
72	0.529	11.221	1.563	0.787	0.499	0.000	0.966	0.579	0.301
73	0.529	11.185	1.247	0.849	0.453	0.000	0.969	0.588	0.302
74	0.529	11.149	0.992	0.910	0.406	0.000	0.999	0.600	0.306
75	1.852	11.252	1.972	0.835	0.488	0.000	1.045	0.618	0.323
76	-4.040	11.063	1.853	0.842	0.494	0.000	0.980	0.553	0.302
77	0.529	11.402	1.336	0.767	0.507	0.000	0.910	0.572	0.297
78	0.529	11.365	1.087	0.807	0.461	0.000	0.910	0.579	0.299
79	0.529	11.328	0.879	0.868	0.413	0.000	0.935	0.587	0.300
80	1.852	11.433	1.652	0.799	0.496	0.000	0.985	0.581	0.307

81	1.852	11.396	1.307	0.862	0.450	0.000	0.988	0.591	0.309
82	1.852	11.359	1.032	0.923	0.403	0.000	1.020	0.607	0.314
83	-4.040	11.240	1.560	0.808	0.501	0.000	0.923	0.544	0.298
84	-4.040	11.204	1.249	0.869	0.455	0.000	0.924	0.549	0.298
85	-4.040	11.169	0.997	0.932	0.408	0.000	0.951	0.554	0.298
86	1.852	11.615	1.408	0.771	0.504	0.000	0.927	0.572	0.303
87	1.852	11.578	1.138	0.821	0.458	0.000	0.928	0.579	0.304
88	1.852	11.541	0.915	0.882	0.410	0.000	0.955	0.588	0.306
89	3.175	11.646	1.746	0.811	0.493	0.000	1.002	0.585	0.315
90	3.175	11.609	1.368	0.874	0.447	0.000	1.007	0.597	0.318
91	3.175	11.571	1.071	0.935	0.400	0.000	1.041	0.625	0.327
92	-4.040	11.419	1.334	0.792	0.509	0.000	0.869	0.541	0.296
93	-4.040	11.383	1.089	0.828	0.463	0.000	0.867	0.546	0.296
94	-4.040	11.347	0.883	0.890	0.416	0.000	0.890	0.551	0.297
95	-2.778	11.451	1.646	0.817	0.498	0.000	0.939	0.545	0.296
96	-2.778	11.415	1.307	0.881	0.452	0.000	0.942	0.550	0.295
97	-2.778	11.379	1.035	0.944	0.405	0.000	0.971	0.556	0.296
98	1.852	11.761	0.997	0.778	0.465	0.000	0.873	0.575	0.303
99	1.852	11.724	0.813	0.838	0.417	0.000	0.896	0.583	0.304
100	3.175	11.831	1.485	0.774	0.500	0.000	0.945	0.573	0.309
101	3.175	11.793	1.191	0.835	0.454	0.000	0.947	0.580	0.310
102	3.175	11.755	0.952	0.896	0.407	0.000	0.976	0.590	0.313
103	4.497	11.862	1.851	0.822	0.490	0.000	1.021	0.591	0.323
104	4.497	11.824	1.435	0.885	0.444	0.000	1.027	0.608	0.327
105	-4.040	11.563	0.955	0.784	0.470	0.000	0.815	0.545	0.296
106	-4.040	11.527	0.785	0.846	0.422	0.000	0.835	0.550	0.297
107	-2.778	11.632	1.404	0.796	0.506	0.000	0.885	0.541	0.293
108	-2.778	11.595	1.139	0.842	0.460	0.000	0.885	0.545	0.293
109	-2.778	11.559	0.919	0.904	0.413	0.000	0.909	0.550	0.294
110	-1.515	11.665	1.742	0.828	0.495	0.000	0.957	0.547	0.294
111	-1.515	11.628	1.369	0.893	0.449	0.000	0.960	0.552	0.293

АНАЛИЗ РЕЗУЛЬТАТОВ

(с указанием номеров вариантов в нисходящей последовательности)

Лучшие варианты относительно передаточного отношения: 4 5 6 8 9 10 17 67 68 72 ...

Лучшие варианты относительно наименьшего веса: 7 3 6 2 16 5 1 15 4 14 ...
 Оптимальные варианты относительно трения (AC/AE): 13 46 91 28 6 61 33 82 16 52 ...
 Лучшие варианты относительно вариации жесткости: 2 3 4 7 9 10 13 14 17 20 ...
 Лучшие варианты относительно прочности: 107 92 83 95 108 105 93 110 84 96 ...
 Лучшие варианты в общем (сумма) : 107 111 108 110 109 96 95 92 97 105 ...

ОПРЕДЕЛЕНИЕ ВАРИАНТОВ ЗУБЧАТОГО ЗАЦЕПЛЕНИЯ

Вводные данные!

Заданное передаточное отношение 3.600 +/- 5.000 % (3.420 ... 3.780)

Мощность: 24.0000 kW

Исходная частота вращения: 1000.00 (Колесо 1)

Модуль 2.5000 mm

Угол зацепления 20.000

Угол наклона линии зуба 23.000

Межосевое расстояние от 125.0000 mm до 135.0000 mm с шагом 1.0000 mm

Смещение исходного контура допустимо в диапазоне -0.500 ... 0.500

Минимальное число зубьев 18

>> Отбрасываются результаты с удельным скольжением выше (zeta > 3.000; zeta < -3.000).

>> Результаты толщины зуба на головке < минимальная толщина зуба отбрасываются.

x1(max)-criteria: согласно скомпенсированному удельному скольжению

>> Момент инерции: Выдаются данные для области делительной окружности до внутреннего диаметра (при внутреннем зацеплении для делительной окружности до наружного диаметра)

>> Варианты, не отвечающие заданным запасам прочности, отклоняются.

Комментарий:

№ = номер варианта

a = Межосевое расстояние

mn = нормальный модуль

alfa = Угол зацепления

бета = угол наклона линии зуба

Z1, Z2 = число зубьев колес 1, 2

X1+X2, X1, X2 = смещение исходного контура, сумма, колесо 1, колесо 2

Ea	= Торцевое перекрытие (eps_a.e+eps_a.i)/2
Eb	= Осевое перекрытие
Eg	= общее перекрытие (eps_g.e+eps_g.i)/2
*	= Варианты, отмеченные *, являются оптимальными (целочисленные общие перекрытия)
слайд	= удельное скольжение (максимальное значение)
i	= передаточное отношение
diff_i	= отклонение от заданного передаточного отношения в %
SF1, SF2	= запас прочности ножки зуба колес 1, 2
SF1, SF2	= запас прочности боковой поверхности (в однопарном зацеплении) колес 1, 2
SB	= запас прочности на заедание, температура «вспышки» в пятне контакта
SInt	= запас прочности на заедание, интегральная температура
del_cg	= Стандартное отклонение жесткости при обкатке (Н/мм/мум)

No.	a	mn	alfa	beta	Z1	Z2	X1+X2	X1	Ea	Eb	Eg	Slide	i	diff_i	SF1					
SF2	SH1	SH2SB			SInt	del_cg														
1	125.00	2.50	20.0	23.0	20	71	+0.591	+0.184	1.40	1.99	3.39*-1.40-0.76	3.55-1.4		2.59	2.57	1.20	1.25	5.22	3.62	0.00
2								+0.284	1.39	1.99	3.38*-1.13-0.84			2.62	2.54	1.20	1.24	5.81	3.63	0.00
3								+0.384	1.37	1.99	3.36*-0.91-0.91			2.64	2.51	1.19	1.24	5.69	3.61	0.00
4	125.00	2.50	20.0	23.0	20	72	+0.027	+0.091	1.48	1.99	3.47*-1.76-0.91	3.60+0.0		2.67	2.69	1.19	1.24	4.79	3.55	0.00
5								+0.191	1.46	1.99	3.45*-1.38-0.99			2.71	2.65	1.18	1.23	5.36	3.56	0.00
6								+0.291	1.44	1.99	3.43*-1.08-1.08			2.74	2.62	1.18	1.22	5.18	3.53	0.00
7	125.00	2.50	20.0	23.0	21	72	-0.498	-0.006	1.55	1.99	3.54*-2.13-1.04	3.43-4.8		2.86	2.93	1.21	1.26	4.62	3.54	0.00
8	126.00	2.50	20.0	23.0	20	72	+0.439	+0.158	1.42	1.99	3.41*-1.48-0.80	3.60+0.0		2.61	2.60	1.20	1.25	5.12	3.61	0.00
9								+0.258	1.41	1.99	3.40*-1.19-0.87			2.64	2.57	1.19	1.24	5.70	3.62	0.00
10								+0.358	1.39	1.99	3.38*-0.95-0.95			2.66	2.54	1.19	1.24	5.56	3.59	0.00
11	126.00	2.50	20.0	23.0	20	73	-0.115	+0.071	1.49	1.99	3.48*-1.87-0.95	3.65+1.4		2.69	2.73	1.19	1.23	4.70	3.54	0.00
12								+0.171	1.48	1.99	3.47*-1.45-1.04			2.74	2.69	1.18	1.23	5.26	3.54	0.00
13								+0.271	1.46	1.99	3.45*-1.13-1.13			2.77	2.65	1.17	1.22	5.06	3.51	0.00
14	126.00	2.50	20.0	23.0	21	72	-0.115	+0.053	1.50	1.99	3.49*-1.75-0.91	3.43-4.8		2.78	2.82	1.23	1.28	4.98	3.61	0.00
15								+0.153	1.49	1.99	3.48*-1.38-1.00			2.82	2.78	1.22	1.27	5.56	3.61	0.00
16								+0.253	1.47	1.99	3.46*-1.09-1.09			2.85	2.73	1.22	1.26	5.38	3.59	0.00
17	127.00	2.50	20.0	23.0	20	72	+0.869	+0.433	1.33	1.99	3.32*-0.84-0.84	3.60+0.0		2.61	2.47	1.19	1.24	5.98	3.65	0.00
18	127.00	2.50	20.0	23.0	20	73	+0.289	+0.133	1.44	1.99	3.43*-1.57-0.83	3.65+1.4		2.63	2.63	1.20	1.25	5.02	3.60	0.00

19											+0.233	1.43	1.99	3.42*-1.25-0.91		2.66	2.59	1.19	1.24	5.60	3.60	0.00
20											+0.333	1.41	1.99	3.40*-0.99-0.99		2.69	2.56	1.19	1.23	5.44	3.58	0.00
21	127.00	2.50	20.0	23.0	20	74	-0.255	+0.052	1.51	1.99	3.50*-1.99-0.99	3.70+2.8	2.72	2.77	1.18	1.23	4.60	3.52	0.00			
22											+0.152	1.49	1.99	3.48*-1.53-1.08		2.77	2.72	1.18	1.22	5.17	3.52	0.00
23	127.00	2.50	20.0	23.0	21	72	+0.289	+0.120	1.45	1.99	3.44*-1.48-0.80	3.43-4.8	2.71	2.71	1.24	1.29	5.32	3.67	0.00			
24											+0.220	1.43	1.99	3.42*-1.19-0.88		2.75	2.68	1.23	1.28	5.92	3.67	0.00
25											+0.320	1.42	1.99	3.41*-0.96-0.96		2.77	2.65	1.23	1.28	5.79	3.65	0.00
26	127.00	2.50	20.0	23.0	21	73	-0.255	+0.032	1.52	1.99	3.51*-1.86-0.95	3.48-3.4	2.81	2.86	1.22	1.27	4.88	3.59	0.00			
27											+0.132	1.50	1.99	3.49*-1.45-1.04		2.85	2.81	1.22	1.27	5.46	3.59	0.00
28											+0.232	1.48	1.99	3.47*-1.13-1.13		2.89	2.77	1.21	1.26	5.26	3.57	0.00
29	128.00	2.50	20.0	23.0	20	73	+0.713	+0.305	1.37	1.99	3.36*-1.08-0.80	3.65+1.4	2.61	2.52	1.20	1.25	5.95	3.65	0.00			
30											+0.405	1.35	1.99	3.34*-0.88-0.88		2.63	2.49	1.19	1.24	5.84	3.63	0.00
31	128.00	2.50	20.0	23.0	20	74	+0.142	+0.111	1.46	1.99	3.45*-1.66-0.87	3.70+2.8	2.65	2.66	1.20	1.25	4.92	3.58	0.00			
32											+0.211	1.45	1.99	3.44*-1.31-0.95		2.69	2.63	1.19	1.24	5.50	3.59	0.00
33											+0.311	1.43	1.99	3.42*-1.03-1.03		2.71	2.59	1.18	1.23	5.32	3.56	0.00
34	128.00	2.50	20.0	23.0	20	75	-0.392	+0.036	1.53	1.99	3.52*-2.12-1.03	3.75+4.2	2.75	2.80	1.18	1.23	4.50	3.51	0.00			
35	128.00	2.50	20.0	23.0	21	72	+0.713	+0.298	1.38	1.99	3.37*-1.04-0.77	3.43-4.8	2.69	2.60	1.24	1.29	6.29	3.72	0.00			
36											+0.398	1.36	1.99	3.35*-0.84-0.84		2.71	2.57	1.24	1.28	6.21	3.71	0.00
37	128.00	2.50	20.0	23.0	21	73	+0.142	+0.096	1.47	1.99	3.46*-1.56-0.83	3.48-3.4	2.74	2.74	1.24	1.29	5.22	3.65	0.00			
38											+0.196	1.45	1.99	3.44*-1.25-0.91		2.77	2.71	1.23	1.28	5.82	3.66	0.00
39											+0.296	1.44	1.99	3.43*-1.00-1.00		2.80	2.68	1.23	1.27	5.66	3.64	0.00
40	128.00	2.50	20.0	23.0	21	74	-0.392	+0.014	1.54	1.99	3.53*-1.98-0.99	3.52-2.1	2.84	2.90	1.22	1.27	4.78	3.57	0.00			
41	129.00	2.50	20.0	23.0	20	74	+0.559	+0.179	1.41	1.99	3.40*-1.41-0.76	3.70+2.8	2.60	2.57	1.21	1.26	5.25	3.64	0.00			
42											+0.279	1.39	1.99	3.38*-1.14-0.84		2.63	2.54	1.20	1.25	5.84	3.64	0.00
43											+0.379	1.38	1.99	3.37*-0.91-0.91		2.65	2.51	1.19	1.24	5.71	3.62	0.00
44	129.00	2.50	20.0	23.0	20	75	-0.002	+0.090	1.48	1.99	3.47*-1.76-0.91	3.75+4.2	2.67	2.70	1.19	1.24	4.83	3.57	0.00			
45											+0.190	1.46	1.99	3.45*-1.38-0.99		2.71	2.66	1.19	1.24	5.41	3.57	0.00
46											+0.290	1.44	1.99	3.43*-1.08-1.08		2.74	2.62	1.18	1.23	5.21	3.54	0.00
47	129.00	2.50	20.0	23.0	21	73	+0.559	+0.169	1.41	1.99	3.40*-1.33-0.73	3.48-3.4	2.68	2.65	1.25	1.30	5.56	3.70	0.00			
48											+0.269	1.40	1.99	3.39*-1.09-0.81		2.71	2.62	1.24	1.29	6.18	3.71	0.00
49											+0.369	1.38	1.99	3.37*-0.88-0.88		2.73	2.59	1.24	1.28	6.07	3.69	0.00
50	129.00	2.50	20.0	23.0	21	74	-0.002	+0.074	1.49	1.99	3.48*-1.65-0.87	3.52-2.1	2.76	2.78	1.24	1.29	5.12	3.64	0.00			
51											+0.174	1.47	1.99	3.46*-1.31-0.95		2.80	2.75	1.23	1.28	5.72	3.64	0.00
52											+0.274	1.45	1.99	3.44*-1.04-1.04		2.83	2.71	1.22	1.27	5.54	3.62	0.00

53	130.00	2.50	20.0	23.0	20	75	+0.407	+0.154	1.43	1.99	3.42*-1.49-0.80	3.75+4.2	2.62	2.60	1.21	1.26	5.15	3.62	0.00
54								+0.254	1.41	1.99	3.40*-1.19-0.87		2.65	2.57	1.20	1.25	5.74	3.63	0.00
55								+0.354	1.40	1.99	3.39*-0.95-0.95		2.67	2.54	1.19	1.24	5.58	3.61	0.00
56	130.00	2.50	20.0	23.0	21	74	+0.407	+0.142	1.44	1.99	3.43*-1.40-0.77	3.52-2.1	2.70	2.68	1.25	1.30	5.46	3.69	0.00
57								+0.242	1.42	1.99	3.41*-1.14-0.84		2.73	2.65	1.24	1.29	6.07	3.70	0.00
58								+0.342	1.40	1.99	3.39*-0.92-0.92		2.75	2.62	1.23	1.28	5.94	3.68	0.00
59	130.00	2.50	20.0	23.0	21	75	-0.144	+0.053	1.51	1.99	3.50*-1.75-0.91	3.57-0.8	2.79	2.82	1.23	1.28	5.02	3.62	0.00
60								+0.153	1.49	1.99	3.48*-1.37-0.99		2.83	2.78	1.23	1.28	5.62	3.63	0.00
61								+0.253	1.47	1.99	3.46*-1.08-1.08		2.86	2.74	1.22	1.27	5.42	3.60	0.00
62	131.00	2.50	20.0	23.0	20	75	+0.836	+0.426	1.34	1.99	3.33*-0.84-0.84	3.75+4.2	2.62	2.48	1.20	1.25	5.99	3.66	0.00
63	131.00	2.50	20.0	23.0	21	74	+0.836	+0.420	1.35	1.99	3.34*-0.81-0.81	3.52-2.1	2.70	2.55	1.24	1.29	6.37	3.73	0.00
64	131.00	2.50	20.0	23.0	21	75	+0.259	+0.118	1.46	1.99	3.45*-1.48-0.80	3.57-0.8	2.72	2.71	1.25	1.30	5.36	3.68	0.00
65								+0.218	1.44	1.99	3.43*-1.19-0.88		2.75	2.68	1.24	1.29	5.97	3.68	0.00
66								+0.318	1.42	1.99	3.41*-0.95-0.95		2.78	2.65	1.23	1.28	5.81	3.66	0.00
67	131.00	2.50	20.0	23.0	21	76	-0.283	+0.034	1.52	1.99	3.51*-1.86-0.94	3.62+0.5	2.81	2.86	1.23	1.28	4.92	3.61	0.00
68								+0.134	1.50	1.99	3.49*-1.44-1.03		2.86	2.82	1.22	1.27	5.52	3.61	0.00
69	132.00	2.50	20.0	23.0	21	75	+0.680	+0.191	1.40	1.99	3.39*-1.27-0.70	3.57-0.8	2.67	2.64	1.25	1.30	5.70	3.73	0.00
70								+0.291	1.39	1.99	3.38*-1.04-0.77		2.70	2.60	1.25	1.30	6.33	3.73	0.00
71								+0.391	1.37	1.99	3.36*-0.85-0.84		2.71	2.57	1.24	1.29	6.23	3.72	0.00
72	132.00	2.50	20.0	23.0	21	76	+0.112	+0.094	1.48	1.99	3.47*-1.56-0.83	3.62+0.5	2.74	2.75	1.24	1.29	5.26	3.66	0.00
73								+0.194	1.46	1.99	3.45*-1.25-0.91		2.78	2.72	1.24	1.29	5.86	3.67	0.00
74								+0.294	1.44	1.99	3.43*-0.99-0.99		2.81	2.68	1.23	1.28	5.69	3.65	0.00
75	132.00	2.50	20.0	23.0	21	77	-0.420	+0.016	1.54	1.99	3.53*-1.97-0.98	3.67+1.9	2.85	2.91	1.23	1.27	4.82	3.59	0.00
76	132.00	2.50	20.0	23.0	22	76	-0.420	-0.005	1.55	1.99	3.54*-1.85-0.95	3.45-4.0	2.93	3.00	1.27	1.32	5.10	3.65	0.00
77	133.00	2.50	20.0	23.0	21	76	+0.527	+0.164	1.42	1.99	3.41*-1.34-0.74	3.62+0.5	2.69	2.66	1.25	1.30	5.60	3.72	0.00
78								+0.264	1.41	1.99	3.40*-1.09-0.81		2.72	2.63	1.25	1.30	6.22	3.72	0.00
79								+0.364	1.39	1.99	3.38*-0.88-0.88		2.73	2.60	1.24	1.29	6.09	3.70	0.00
80	133.00	2.50	20.0	23.0	21	77	-0.032	+0.073	1.49	1.99	3.48*-1.65-0.87	3.67+1.9	2.77	2.79	1.24	1.29	5.16	3.65	0.00
81								+0.173	1.48	1.99	3.47*-1.31-0.95		2.81	2.75	1.23	1.28	5.76	3.66	0.00
82								+0.273	1.46	1.99	3.45*-1.03-1.03		2.84	2.71	1.23	1.28	5.57	3.63	0.00
83	133.00	2.50	20.0	23.0	22	76	-0.032	+0.057	1.50	1.99	3.49*-1.56-0.83	3.45-4.0	2.85	2.87	1.28	1.33	5.45	3.71	0.00
84								+0.157	1.48	1.99	3.47*-1.25-0.91		2.89	2.83	1.28	1.33	6.07	3.72	0.00
85								+0.257	1.47	1.99	3.46*-1.00-1.00		2.92	2.80	1.27	1.32	5.90	3.70	0.00

86	134.00	2.50	20.0	23.0	21	77	+0.376	+0.139	1.44	1.99	3.43*-1.41-0.77	3.67+1.9	2.71	2.69	1.25	1.30	5.49	3.70	0.00
87								+0.239	1.43	1.99	3.42*-1.14-0.84		2.74	2.66	1.25	1.30	6.11	3.71	0.00
88								+0.339	1.41	1.99	3.40*-0.91-0.91		2.76	2.63	1.24	1.29	5.96	3.69	0.00
89	134.00	2.50	20.0	23.0	21	78	-0.173	+0.053	1.51	1.99	3.50*-1.75-0.90	3.71+3.2	2.79	2.83	1.24	1.29	5.06	3.64	0.00
90								+0.153	1.49	1.99	3.48*-1.37-0.99		2.84	2.79	1.23	1.28	5.67	3.64	0.00
91								+0.253	1.47	1.99	3.46*-1.07-1.07		2.87	2.75	1.22	1.27	5.46	3.61	0.00
92	134.00	2.50	20.0	23.0	22	76	+0.376	+0.128	1.45	1.99	3.44*-1.33-0.74	3.45-4.0	2.79	2.77	1.29	1.34	5.80	3.77	0.00
93								+0.228	1.43	1.99	3.42*-1.09-0.81		2.82	2.73	1.29	1.34	6.44	3.77	0.00
94								+0.328	1.42	1.99	3.41*-0.88-0.88		2.84	2.70	1.28	1.33	6.32	3.76	0.00
95	134.00	2.50	20.0	23.0	22	77	-0.173	+0.035	1.52	1.99	3.51*-1.65-0.87	3.50-2.8	2.88	2.91	1.28	1.33	5.35	3.70	0.00
96								+0.135	1.50	1.99	3.49*-1.31-0.95		2.92	2.87	1.27	1.32	5.97	3.71	0.00
97								+0.235	1.48	1.99	3.47*-1.03-1.03		2.95	2.83	1.27	1.32	5.78	3.68	0.00
98	135.00	2.50	20.0	23.0	21	77	+0.802	+0.313	1.37	1.99	3.36*-1.00-0.75	3.67+1.9	2.69	2.59	1.25	1.30	6.47	3.76	0.00
99								+0.413	1.36	1.99	3.35*-0.81-0.81		2.70	2.56	1.25	1.30	6.38	3.74	0.00
100	135.00	2.50	20.0	23.0	21	78	+0.228	+0.115	1.46	1.99	3.45*-1.48-0.80	3.71+3.2	2.73	2.72	1.25	1.30	5.39	3.69	0.00
101								+0.215	1.45	1.99	3.44*-1.19-0.87		2.76	2.69	1.24	1.29	6.01	3.70	0.00
102								+0.315	1.43	1.99	3.42*-0.95-0.95		2.78	2.66	1.24	1.29	5.84	3.68	0.00
103	135.00	2.50	20.0	23.0	21	79	-0.312	+0.035	1.53	1.99	3.52*-1.85-0.94	3.76+4.5	2.82	2.87	1.23	1.29	4.97	3.62	0.00
104								+0.135	1.51	1.99	3.50*-1.43-1.02		2.87	2.83	1.23	1.28	5.57	3.62	0.00
105	135.00	2.50	20.0	23.0	22	76	+0.802	+0.307	1.38	1.99	3.37*-0.96-0.72	3.45-4.0	2.76	2.66	1.29	1.34	6.82	3.82	0.00
106								+0.407	1.36	1.99	3.35*-0.78-0.78		2.78	2.63	1.29	1.34	6.76	3.80	0.00
107	135.00	2.50	20.0	23.0	22	77	+0.228	+0.102	1.47	1.99	3.46*-1.40-0.77	3.50-2.8	2.81	2.80	1.29	1.34	5.70	3.76	0.00
108								+0.202	1.45	1.99	3.44*-1.14-0.84		2.84	2.77	1.29	1.34	6.33	3.76	0.00
109								+0.302	1.44	1.99	3.43*-0.92-0.92		2.86	2.74	1.28	1.33	6.19	3.74	0.00
110	135.00	2.50	20.0	23.0	22	78	-0.312	+0.015	1.53	1.99	3.52*-1.74-0.90	3.55-1.5	2.91	2.96	1.28	1.33	5.25	3.69	0.00
111								+0.115	1.52	1.99	3.51*-1.37-0.99		2.95	2.91	1.27	1.32	5.87	3.69	0.00

конец протокола

lines: 324